

JIS

JAPANESE INDUSTRIAL STANDARD

**Methods of measuring dust
concentration in flue gas**

JIS Z 8808—1995

Translated and Published

by

Japanese Standards Association

In the event of any doubt arising,
the original Standard in Japanese is to be final authority.

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JAPANESE INDUSTRIAL STANDARD

J I S

Methods of measuring dust
concentration in flue gas

Z 8808-1995

1. Scope This Japanese Industrial Standard specifies the methods of measuring dust concentration in flue gas, in a flue, chimney, duct, or the like (hereafter referred to as "duct").

Remarks 1. The standards cited in this Standard are shown in Attached Table 1.

2. The units and numerical values in { } in this Standard are based on the traditional units and are appended for informative reference.

2. Definitions For the main terms used in this Standard, the definitions in JIS Z 8103 apply, and the rest of the terms are as follows.

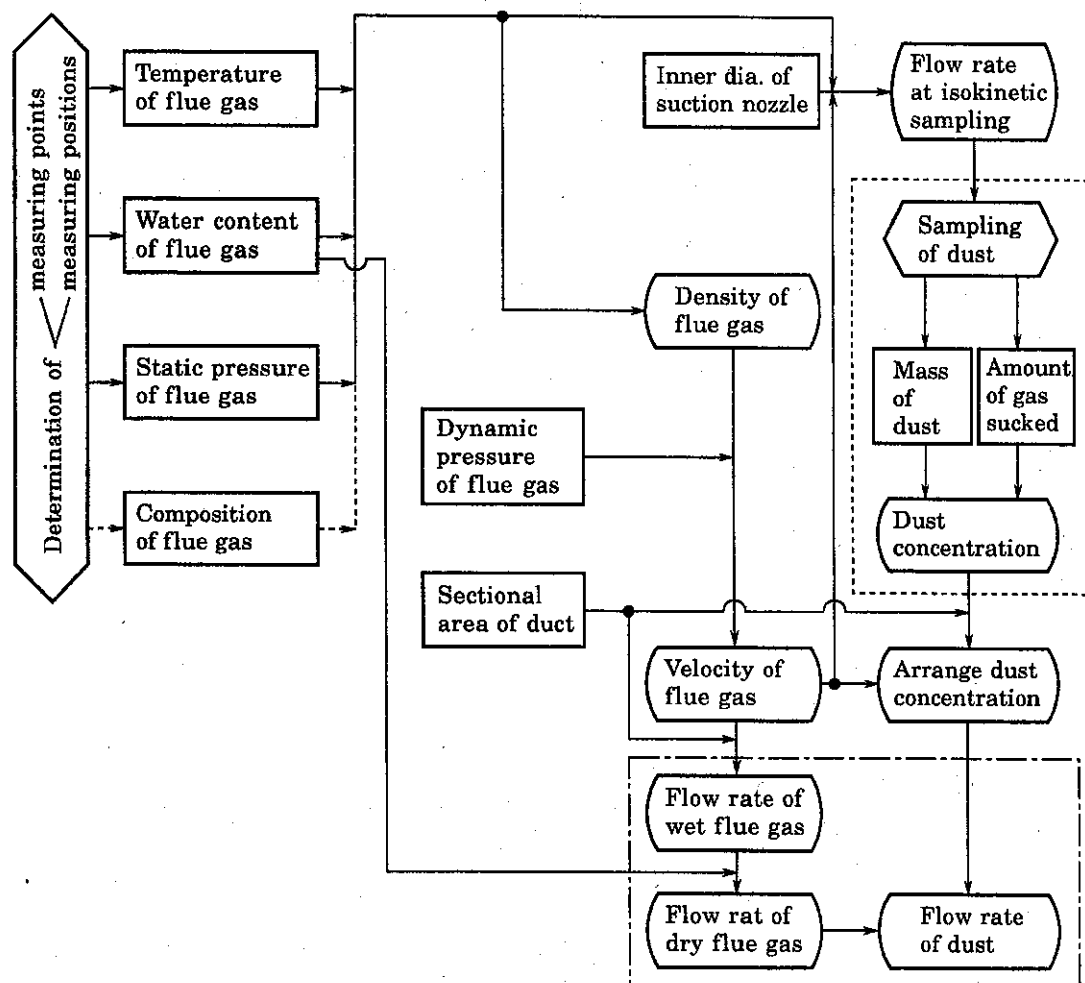
- (1) flue gas The gas containing the solid particles produced during the combustion, synthesis, and decomposition of the fuel and others, during the use of electricity for heat source and mechanical work and others. The gas containing no moisture is called dry flue gas, and the gas containing moisture, wet flue gas.
- (2) dust Solid particles such as smoke dust and powdery dust from which adhesive moisture has been removed by drying. However, soluble substance or the like contained in mist (liquid particle), which is measured by this testing method, shall be regarded as the dust.
- (3) dust concentration The mass of dust that is contained in 1 m³ of dry flue gas at standard condition [temperature 0°C, atmospheric pressure 101.3 kPa (760 mmHg)]. g/m³_N is used as the unit.
Further, m³_N is used as the unit of gas volume in the standard condition.
- (4) moisture content in flue gas The ratio of water vapor contained in flue gas. It is expressed by volumetric percentage (%).
- (5) isokinetic sampling When the flue gas is sampled by using a suction nozzle to prepare a dust sample, the flue gas shall be sucked with the same flow velocity as that of the flue gas under the condition that the suction nozzle is confronted with the flow of the flue gas.
- (6) suction nozzle The nozzle used for suction of flue gas by inserting it into a duct to take a dust sample.
- (7) sampling apparatus for dust The apparatus to collect the dust sample by filtering, while sucking the flue gas with isokinetic sampling.

There are two types of the ordinary-type sampling apparatus and the balanced-type sampling apparatus.

3. Summary of measurement Provide measuring holes at the measuring position of duct selected in accordance with the specifications in 4., and determine the measuring points in accordance with the specifications in the section of the duct including the measuring holes. Insert the suction nozzles of a sampling apparatus for dust into the duct through the measuring holes and place the tips of the nozzles to meet the measuring points. Aspirate the flue gas according to the isokinetic sampling, and obtain the dust concentration by calculation from the quantity of the dust collected by filtering in the dust collector and the volume of gas sucked in it concurrently.

The measuring method is outlined in Fig. 1

Fig. 1. Outline of measuring method



- Remarks
1. The item in $\square$ shows the measured value and that in $\square$ the calculated one.
 2. The processes indicated in the dashed line are the measurements carried out by the balanced-type sampling apparatus.
 3. The processes indicated in chain line are needed for measuring flow rate of flue gas or flow rate of dust.

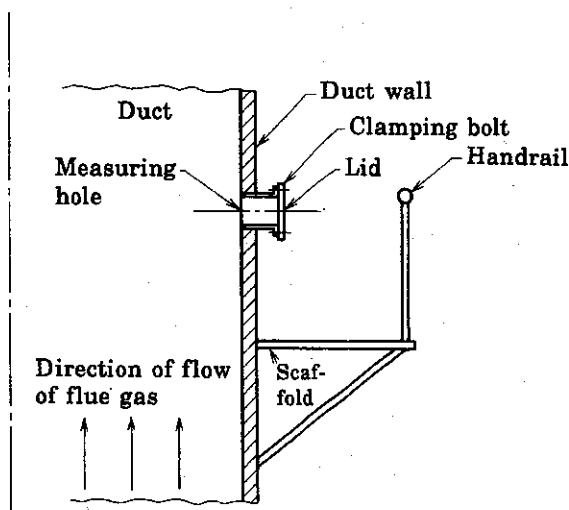
4. Measuring position, hole, and point

4.1 Measuring position The measuring position shall be located at a place where the flue gas is under a comparatively uniform flow and the measurement is performed easily and safely, avoiding the place where a duct bends or its sectional shape is sharply changed.

4.2 Measuring hole The measuring hole of about 100 mm to 150 mm in inner diameter shall be provided on the wall of duct at the measuring position as shown in Fig. 2, and it shall be covered tightly with a suitable lid except measuring time. The positions of measuring holes shall be determined on the perpendicularly-crossed two diameters including all the measuring points, in the case of the duct of circular section, as shown in Fig. 3; and on the prolonged straight lines, vertically or laterally, including all measuring points, in the case of the duct with rectangular or square type section, as shown respectively in Fig. 4 and Fig. 5. The measuring holes for the duct having other sectional shape than above-mentioned, shall be decided similarly to the above method.

The opening between the inserted device and the measuring hole shall be packed closely with such material as heat resisting material.

Fig. 2. Example of structure of measuring hole



4.3 Measuring point The section of a duct chosen as the measuring position shall be equally divided into a suitable number of areas according to its shape and size as described below, and the measuring point shall be allotted in each area.

When the deposits of dust and others exist around the section for measurement, the section for measurement shall be determined excepting such area.

- (1) In case of circular section Measuring points shall be selected at the positions given in Table 1 on the diameters intersecting at right angles to each other in the measuring section as shown in Fig. 3. The number of measuring points shall be up to 20 when the diameter of the duct is over 4.5 m.

Fig. 3. Example of measuring points on circular section

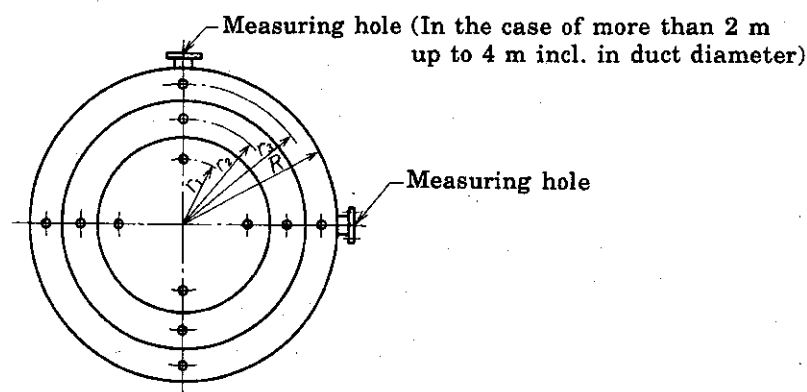


Table 1. Measuring points on circular section

Applicable diameter of duct $2R$ (m)	Number of divisions by radius	Number of measuring points	Distance from center of duct to measuring point (m)				
			r_1	r_2	r_3	r_4	r_5
1 and under	1	4	$0.707R$	—	—	—	—
Over 1 to 2 incl.	2	8	$0.500R$	$0.866R$	—	—	—
Over 2 to 4 incl.	3	12	$0.408R$	$0.707R$	$0.913R$	—	—
Over 4 to 4.5 incl.	4	16	$0.354R$	$0.612R$	$0.791R$	$0.935R$	—
Over 4.5	5	20	$0.316R$	$0.548R$	$0.707R$	$0.837R$	$0.949R$

- (2) In the case of rectangular and square sections As shown in Figs. 4 and 5, the measuring section shall be divided into four or more rectangles or squares of equal sectional area of which one side length (l) is limited within 1 m, and the measuring points shall be allotted at their centers. The applicable dimension and the decision of the measuring points shall be as shown in Table 2. When the sectional area of the duct exceeds 20 m², the measuring points shall be, generally, at most 20 points and each of which has an equal sectional area.

When the flow in the measuring section is a symmetry, the length of one side of a divided section in the direction of a symmetry shall be made smaller than that of a side perpendicular to it, and the number of measuring points shall increase respectively.

Fig. 4. Example of measuring points on the rectangular section (In the case of 12 measuring points)

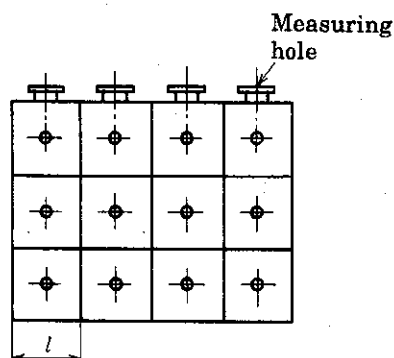


Fig. 5. Example of measuring points on the square section (in the case of 16 measuring points)

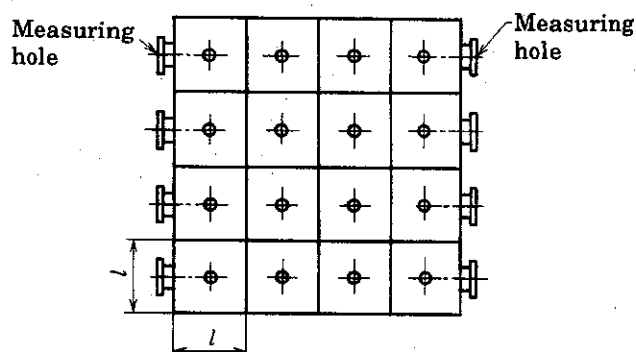


Table 2. Allotment of measuring points on the rectangular and square sections

Sectional area of applicable duct $A \text{ (m}^2\text{)}$	Length of a divided side $l \text{ (m)}$
1 and under	$l \leq 0.5$
Over 1 to 4 incl.	$l \leq 0.667$
Over 4 to 20 incl.	$l \leq 1$

- (3) In the case of section with other forms The measuring points shall be selected according to (1) or (2).

- Remarks
1. In the case of a small duct (0.25 m^2 or less of sectional area), it is permissible to choose the center of the section as a measuring point.
 2. When the distribution of flow velocity in the measuring section can be regarded as comparatively symmetry according to the specification in 7.3, the number of measuring points may be reduced to 1/2 and 1/4 respectively by employing one side on the vertical symmetry axis in the case of a horizontal duct, and employing quarter part in the case of a vertical duct.
 3. As far as the distribution of dust concentration measured on the base of this Standard has been known, and it has been confirmed that the average dust concentration can be calculated with the measurement of one or several points, these points may be taken as the representative for the measurements.

5. Measurement of temperature of flue gas

5.1 Measuring point Select the measuring points according to the specifications of 4.3. However, the number of measuring points may be reduced when temperature distribution is confirmed to be comparatively flat.

5.2 Measuring apparatus Employ the thermometer specified in JIS B 7413, JIS C 1601, or JIS C 1603.

5.3 Measuring method Insert the measuring apparatus into the measuring hole to meet the measuring point. Measure temperature in accordance with JIS Z 8704 or JIS Z 8705.

The average of measurements at measuring points shall be the temperature of flue gas.

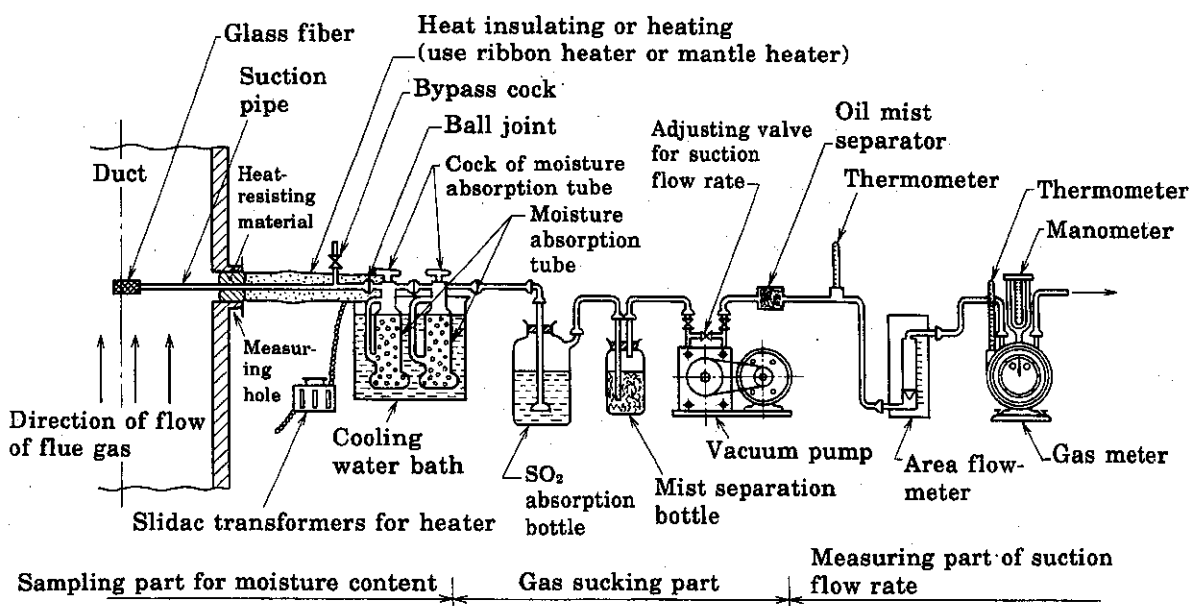
6. Measurement of moisture content in flue gas

6.1 Method by moisture absorption tube

6.1.1 Measuring point Select the measuring point near the center of the section of duct, in the measuring position specified in 4.1.

6.1.2 Sampling apparatus for moisture content The sampling apparatus for moisture content shall be composed of the sampling part for moisture content, gas suction part, and measuring part of suction flow rate as described below. An example of arrangement is shown in Fig. 6. There shall be no gas leakage at every connecting part of this sampling apparatus.

Fig. 6. Example of arrangement of sampling apparatus for moisture content



(1) Sampling part for moisture content The sampling part for moisture content shall be composed of suction tube, moisture absorption tube, and cooling water bath as follows:

(a) Suction tube Made of hard glass or metal, packed with glass fiber or the like at its end so as to prevent the mixing of dust when flue gas is sucked.

- (b) Moisture absorption tube U-tube specified in JIS R 3503 or an absorption tube specified in JIS M 8813 filled with moisture absorbent such as calcium chloride specified in JIS K 8125, and packed with the glass fiber to prevent the moisture absorbent from scattering. Generally, use two moisture absorption tubes.
 - (c) Cooling water bath Use for cooling the moisture absorption tubes.
- (2) Gas sucking part The gas sucking part shall be composed of a SO₂ absorption bottle, mist separation bottle, suction apparatus, and adjusting valve for suction flow rate as follows:
- (a) SO₂ absorption bottle Prepare absorption liquid (about 3%) by diluting about ten times the hydrogen peroxide solution specified in JIS K 8230. Use the bottle if it is feared that suction apparatus may be corroded by sulfur oxide and others.
 - (b) Mist separation bottle This bottle, packed with glass fiber or absorbent cotton, can be used to prevent the entrainment of SO₂ absorption liquid.
 - (c) Suction apparatus The apparatus for sucking flue gas, such as a vacuum pump or blower, which is strong enough to endure the negative pressure in duct and the resistance of each part of the sampling apparatus and capable of sucking gas at a required flow rate. At discharge side of the suction appliance, if required, an oil mist separator is attached to protect a flowmeter.
 - (d) Adjusting valve for suction flow rate The valve is for adjusting suction flow rate of flue gas, and is installed at suitable place of the suction apparatus or gas sucking part.
- (3) Measuring part of suction flow rate The measuring part of suction flow rate shall be composed of an integrating flowmeter, the area flowmeter that has been calibrated by the integrating flowmeter, and the instantaneous flowmeter such as contraction device, as described below. The integrating flowmeter is used for measuring the volume of gas sucked in, and the instantaneous flowmeter, for confirmation of suction flow rate.
- (a) Integrating flowmeter A gas meter either of wet type or of dry type is used. In the case of dry type gas meter, the gas shall be dried prior to introducing.
 - (b) Area flowmeter or contraction device The area flowmeter specified in JIS B 7551 or the contraction device specified in JIS Z 8762 shall be used.

The measurement by the area flowmeter shall be in accordance with the specifications in JIS Z 8761. The measurement by the contraction device shall be in accordance with that in JIS Z 8762.

Remarks: In the case of the measurement of moisture content, the instantaneous flowmeter can be omitted.

6.1.3 Measuring method Measuring method shall be as follows:

(1) Preparation of measurement

- (a) Handling of moisture absorption tube After filling the tube with moisture absorbent, wipe the extraneous matters off from its surface, then tightly close its cock, and weigh it to obtain the mass (m_{a1}).
- (b) Heat insulation of suction tube Heat insulation or heating shall be made for the suction tube so as to prevent the inside of the tube from condensation of moisture.
- (c) Cooling of moisture absorption tube When the temperature of flue gas is higher than the atmospheric temperature, employ a cooling water bath shown in Fig. 6.

(2) Measurement

- (a) Determination of suction flow rate Control the suction flow rate of flue gas with adjusting valve for suction flow rate so that it may be 0.1 l/min or less per 1 g of moisture absorbent in one moisture absorption tube. In order to confirm the suction flow rate, use the instantaneous flowmeter or integrating flowmeter.
- (b) Sampling for moisture content After inserting a suction tube into a measuring hole and confirming the heat insulating or heating, replace fully the air kept in the suction tube by flue gas by using the bypass cock shown in Fig. 6, and then shut the bypass cock and commence the suction of the flue gas by opening the inlet and outlet cocks of the moisture absorption tubes. Select the volume of suction gas so that the absorbed moisture may range from 0.1g to 1 g. During the suction, measure the temperature and pressure of suction gas in the gas meter, and the volume of sucked gas. After the required volume of flue gas is sucked, close the cocks of moisture absorption tubes and take off all piping adjacent to the moisture absorption tubes.

Measure the volume of suction gas by using an integrating flowmeter and read the value to the nearest 0.1 l.

- (c) Weighing After fully removing the moisture and extraneous matters on the surface of the moisture absorption tubes, weigh and obtain the mass (m_{a2}). Employ a balance having the reciprocal sensibility of not more than 10 mg.

(3) Calculation of moisture content The calculation of moisture content in the flue gas shall be carried out as follows:

- (a) In the case of employing a wet gas meter In the case of employing a wet gas meter, calculate according to the following formula (1).

$$x_w = \frac{\frac{22.4}{18} m_a}{V_m \times \frac{273}{273 + \theta_m} \times \frac{P_a + P_m - P_v}{101.3} + \frac{22.4}{18} m_a} \times 100 \dots\dots\dots (1)$$

where, x_w : volume percentage of water vapor in flue gas (%)
 m_a : mass of moisture absorbed ($m_{a2} - m_{a1}$) (g)
 V_m : volume of gas sucked (reading of gas meter) (l)
 θ_m : temperature of suction gas in gas meter ($^{\circ}\text{C}$)
 P_a : atmospheric pressure (kPa){mmHg}
 P_m : gauge pressure of gas in the gas meter (kPa){mmHg}
 P_v : saturated water vapor pressure at θ_m (kPa){mmHg}

Informative reference: When using mmHg as the unit, use the value of 760 instead of 101.3 in formula (1).

- (b) In the case of employing a dry gas meter In this case, calculate by using formula (1) after eliminating a term P_v and reading V_m as sucked gas volume (reading of a gas meter). However, in this case, the gas should be dried before the dry gas meter.

6.2 Method by calculation The moisture content in flue gas can be calculated according to the following formula (2).

$$x_w = \frac{w_g}{G} \times \frac{22.4}{18} \times 100 \dots\dots\dots (2)$$

where, x_w : volume percentage of water vapor in flue gas (%)
 G : amount of wet flue gas per unit amount of fuel [$\text{m}^3_{\text{N}}/\text{kg}$ (solid or liquid fuel), $\text{m}^3/\text{m}^3_{\text{N}}$ (gaseous fuel)]
 w_g : moisture content in generated gas per unit amount of fuel [kg/kg (solid or liquid fuel), $\text{kg}/\text{m}^3_{\text{N}}$ (gaseous fuel)]

For solid and liquid fuels:

$$w_g = \frac{29}{22.4} A_v x + \frac{w_r}{100} + \frac{9h}{100}$$

For gaseous fuel:

$$w_g = \frac{29}{22.4} A_v x + \frac{18}{22.4} \times \frac{1}{100} (H_2 + 2CH_4 + 2C_2H_4 + \dots\dots)$$

where, A_v : volume of dry air consumed per unit amount of fuel [$\text{m}^3_{\text{N}}/\text{kg}$ (solid or liquid fuel), $\text{m}^3/\text{m}^3_{\text{N}}$ (gaseous fuel)]

x : absolute humidity of air for combustion [mass ratio of the amount of water vapor in wet air to dry air (kg/kg dry air)]

$$x = \frac{0.622\phi P_v}{100P_a - \phi P_v}$$

where, ϕ : relative humidity (%)

P_v : saturated water vapor pressure (kPa) (mmHg)

P_a : atmospheric pressure (kPa)(mmHg)

w_r : mass percentage of total moisture content in fuel (%)

h : mass percentage of hydrogen (at the base of using time) contained in fuel (%)

H_2 : volume percentage of hydrogen in fuel gas (%)

CH_4 : volume percentage of methane in fuel gas (%)

C_2H_4 : volume percentage of ethylene in fuel gas (%)

Remarks 1. When moisture generated from the steam blowing, product and others, has been contained in the flue gas, the moisture content shall be added to w_g after being converted into that per unit amount of fuel according to the following.

(1) For solid or liquid fuel:

$$G = G' + \frac{22.4}{18} w_g \dots\dots\dots (3)$$

In this case,

$$G' = (m - 0.21)A_o + 1.867 \frac{c'}{100} + 0.7 \frac{s}{100} + 0.8 \frac{n}{100}$$

$$m = \frac{(N_2)}{(N_2) - 3.76[(O_2) - 0.5(CO)]}$$

$$A_o = \frac{1}{100} [8.89c' + 26.7(h - \frac{o}{8}) + 3.33s]$$

or

$$G' = \frac{1.867c' + 0.7s}{(CO_2) + (CO)}$$

where, G' : amount of dry flue gas per unit amount of fuel ($\text{m}^3_{\text{N}}/\text{kg}$)

m : air ratio

w : mass percentage of total moisture contained in fuel (%)

h : mass percentage of hydrogen contained in fuel (%)

c' : mass percentage of combustible carbon contained in fuel (%)

s : mass percentage of combustible sulfur contained in fuel (%)

n : mass percentage of nitrogen contained in fuel (%)

o : mass percentage of oxygen contained in fuel (%)

A_0 : theoretical volume of air per 1 kg of fuel ($\text{m}^3_{\text{N}}/\text{kg}$ fuel)

(N_2) : volume percentage of nitrogen contained in dry flue gas (%)

(O_2) : volume percentage of oxygen contained in dry flue gas (%)

(CO) : volume percentage of carbon monoxide contained in dry flue gas (%)

(CO_2) : volume percentage of carbon dioxide contained in dry flue gas (%)

(2) For gaseous fuel:

$$G = G' + \frac{1}{100} (H_2 + 2CH_4 + 2C_2H_4 + \dots) \dots\dots\dots (4)$$

In this case,

$$G' = \frac{(CO + CO_2 + CH_4 + 2C_2H_4 + \dots)}{(CO_2) + (CO)}$$

where, G' : amount of dry flue gas per 1 m^3_{N} of fuel gas ($\text{m}^3_{\text{N}}/\text{m}^3_{\text{N}}$)

H_2 : volume percentage of hydrogen in dry fuel gas (%)

CH_4 : volume percentage of methane in dry fuel gas (%)

C_2H_4 : volume percentage of ethylene in dry fuel gas (%)

CO : volume percentage of carbon monoxide in dry fuel gas (%)

CO_2 : volume percentage of carbon dioxide in dry fuel gas (%)

Remarks 2. When mist coexists in the flue gas at the outlet of scrubber or others, the moisture content can be calculated according to the following formula (5), using the saturated water vapor pressure at the temperature of flue gas (in the case at 100°C or under).

$$x_w = \frac{P_v}{P_a + P_s} \times 100 \dots\dots\dots (5)$$

where, x_w : Volume percentage of water vapor in flue gas (%)
 P_v : saturated water vapor pressure at temperature of flue gas (kPa){mmHg}
 P_a : atmospheric pressure (kPa){mmHg}
 P_s : static pressure of flue gas (kPa){mmHg}

7. Measurement of flow velocity and flow rate of flue gas

7.1 Measuring point Select the measuring point in accordance with the specification in 4.3.

7.2 Measuring apparatus Measuring apparatus shall be as follows:

(1) Pitot tube Use a L-type Pitot tube specified in JIS B 8330, or the Pitot tube of which Pitot tube coefficient has been known.

Remarks: For measuring the flow velocity of flue gas, the flowmeter for gas, such as the anemometer specified in JIS T 8202 calibrated by a Pitot tube, or Karman vortex flowmeter, may be used. When an indicated value is influenced, however, by the characteristic of flue gas (temperature, pressure, and composition) and the property of dust, correct them by using the value measured by a Pitot tube.

(2) Pressure gauge For pressure gauge, micromanometer⁽¹⁾ such as an inclined-tube manometer or others, U-tube manometer, or a gauge capable of converting the pressure into a quantity of electricity and indicating, shall be used.

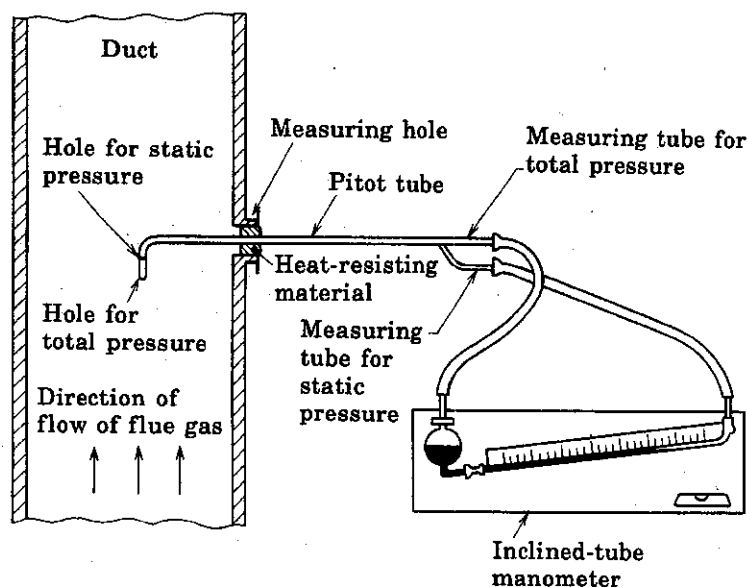
Note ⁽¹⁾ The ratio of liquid surface areas between the water tubes of micromanometer such as an inclined-tube manometer and the water tank shall be not less than 100 times.

7.3 Measuring method of flow velocity of flue gas

7.3.1 Measuring method of dynamic pressure of flue gas Insert a Pitot tube through a measuring hole to the measuring point in a duct as shown in Fig. 7, let the total pressure hole be confronted properly with the flow direction of flue gas, and measure the dynamic pressure by means of a pressure gauge. Principally, read the value of dynamic pressure to the nearest 1 Pa {0.1 mmH₂O}.

In this case, the angular deviation of the Pitot tube to the flow direction of flue gas shall be 10° or less.

Fig. 7 Example for measurement of flow velocity of flue gas with a Pitot tube



7.3.2 Measuring method of static pressure of flue gas At each measuring point, measure the pressure at static pressure hole of a Pitot tube installed similarly to 7.3.1, by using a pressure gauge. However, the number of measuring points may be reduced.

Remarks: When a static pressure hole is provided on a duct, the measurement by using this hole is permissible.

7.3.3 Calculation of density of flue gas Obtain it through calculation using the composition of flue gas, or calculate it in accordance with the following formula (6) using measured value by a gas density meter.

$$\rho = \rho_0 \times \frac{273}{273 + \theta_s} \times \frac{P_a + P_s}{101.3} \dots\dots\dots (6)$$

$$\left\{ \gamma = \gamma_0 \times \frac{273}{273 + \theta_s} \times \frac{P_a + P_s}{760} \right\}$$

In this case,

$$\rho_0 = \frac{1}{22.4 \times 100} [(M_1 x_1 + M_2 x_2 + \dots + M_n x_n) (1 - \frac{x_w}{100}) + 18 x_w]$$

$$\{\gamma_0 = \frac{1}{22.4 \times 100} [M_1 x_1 + M_2 x_2 + \dots + M_n x_n) (1 - \frac{x_w}{100}) + 18 x_w]\}$$

or

$$\rho_0 = \rho' (1 - \frac{x_w}{100}) + \frac{18}{22.4 \times 100} x_w$$

$$\left\{ \gamma_0 = \gamma' (1 - \frac{x_w}{100}) + \frac{18}{22.4 \times 100} x_w \right\}$$

- where,
- ρ : density of flue gas in duct (kg/m^3)
 - $\{\gamma$: weight per unit volume of flue gas in duct ($\text{kgf/m}^3\}$
 - ρ_0 : density of wet flue gas converted into that at 0°C in temperature and 101.3 kPa in atmospheric pressure (kgf/m^3)
 - $\{\gamma_0$: weight per unit volume of wet flue gas converted into that at 0°C in temperature and 760 mmHg in atmospheric pressure (kgf/m^3)
 - $\rho^{(2)}$: density of dry flue gas (kg/m^3)
 - $\{\gamma^{(2)}$: weight per unit volume of dry flue gas (kgf/m^3)
 - P_a : atmospheric pressure (kPa){mmHg}
 - P_s : average value of static pressure of flue gas obtained at each measuring point (kPa){mmHg}
 - θ_s : average value of temperature of flue gas obtained at each measuring point ($^\circ\text{C}$)
 - $M_1, M_2, \dots, M_n$: molecular weight of each component in flue gas
 - $x_1, x_2, \dots, x_n$: volume percentage of each component in dry flue gas (%) (refer to JIS K 2301)
 - x_w : volume percentage of water vapor in flue gas (%)

Note (2) These values can be obtained by using a gas density meter.

Informative reference: Generally, when the solid fuel and liquid fuel are burnt with air, the relation $\rho_0 = 1.30 \text{ kg/m}^3$ $\{\gamma_0 = 1.30 \text{ kgf/m}^3\}$ is admitted.

7.3.4 Calculation of flow velocity of flue gas Calculate the flow velocity of flue gas at each measuring point according to following formula (7):

$$v = c \sqrt{\frac{2P_d}{\rho}} \dots\dots\dots (7)$$

$$\left\{ v = c \sqrt{\frac{2gh}{\gamma}} \right\}$$

- where,
- v : flow velocity (m/s)
 - c : Pitot tube coefficient
 - P_d : value of dynamic pressure measured by Pitot tube (Pa)
 - ρ : density of flue gas in duct (kg/m^3)
 - $\{h$: value of dynamic pressure measured by Pitot tube (kgf/m^2 or $\text{mmH}_2\text{O}\}$

γ : weight per unit volume of flue gas in duct
(kgf/m³)

g : gravitational acceleration = 9.81 m/s²

7.4 Calculation of flow rate of flue gas

7.4.1 Calculation of flow rate of wet flue gas The flow rate of wet flue gas shall be calculated as follows.

- (1) In the case of calculation from flow velocity of flue gas and sectional area of duct In this case, the flow rate shall be obtained according to the following formula (8):

$$Q_N = A\bar{v} \times \frac{273}{273 + \theta_s} \times \frac{P_a + P_s}{101.3} \times 60 \times 60 \dots\dots\dots (8)$$

where, Q_N : flow rate of wet flue gas (m³/h)

A : sectional area of duct (m²)

P_a : atmospheric pressure (kPa){mmHg}

P_s : average value of static pressure of flue gas at each measuring point (kPa){mmHg}

θ_s : average value of temperature of flue gas at each measuring point (°C)

$\bar{v}$: average value of flow velocity at each measuring point (m/s)

$$\bar{v} = \frac{1}{n}(v_1 + v_2 + \dots + v_n)$$

where, n : number of measuring points

$v_1, v_2, \dots, v_n$: flow velocity at each measuring point (m/s)

Informative reference: When using mmHg as the unit, adopt 760 instead of 101.3 in the formula (8).

- (2) In the case of combustion calculation In the case of combustion calculation, the flow rate shall be obtained according to the following formula (9):

$$Q_N = GW \dots\dots\dots (9)$$

where, Q_N : flow rate of wet flue gas (m³/h)

G : amount of wet flue gas per unit amount of fuel [m³/kg (solid or liquid fuel), m³/m³ (gaseous fuel)] (Refer to 6.2)

W : consumed amount of fuel per 1 h [kg/h (solid or liquid fuel), m³/h (gaseous fuel)]

7.4.2 Calculation of flow rate of dry flue gas The flow rate of dry flue gas shall be calculated according to following formula (10) on the base of flow rate of wet flue gas obtained in 7.4.1.

$$Q'_N = Q_N \left(1 - \frac{x_w}{100}\right) \dots\dots\dots (10)$$

where, Q'_N : flow rate of dry flue gas (m^3/h)
 Q_N : flow rate of wet flue gas (m^3/h)
 x_w : volume percentage of water vapor in flue gas (%)

8. Sampling apparatus for dust

8.1 Kind of sampling apparatus for dust Sampling apparatus for dust shall be classified into the following two kinds according to its principle, construction and function.

- (1) Ordinary-type sampling apparatus The ordinary-type sampling apparatus measures preliminarily various conditions (temperature, pressure, moisture content, density, flow velocity, etc.) of flue gas and suction gas at measuring points to obtain isokinetic sampling flow rate, and sucks flue gas according to the flow rate to filtrate sampling of dust. The apparatus is classified into of manual or automatic driven according to its structure and function.
- (2) Balanced-type sampling apparatus The balanced-type sampling apparatus carries out filtrate sampling of dust directly by isokinetic sampling without obtaining isokinetic sampling flow rate beforehand, having the dynamic or static pressure equilibrated with the pressure of suction gas. The apparatus is classified into of manual or automatic driven according to its structure and function.

8.2 Arrangement of the sampling apparatus for dust The sampling apparatus for dust shall be basically composed of a dust collection part, a gas suction part, and a suction flow rate measuring part.

The sampling apparatus for dust shall be free from leak of gas at all joint parts.

Further, it is classified into type 1 and type 2 according to the position of the dust collector of the sampling apparatus. Type 1 shall place the dust collector in the duct, and type 2 shall place the dust collector outside the duct. In the case of type 2, the dust collecting part shall be kept warm or heated to the level that temperature of flue gas does not lower to dew point.

Remarks: In analysis of heavy metals or the like contained in flue gas, for sampling in the case of containing volatile dust gasified at a temperature of flue gas, a gas absorption part shall be installed additionally between the dust collection part of a sampling apparatus and its gas suction part in order to collect flue gas in the absorbing liquid by cooling.

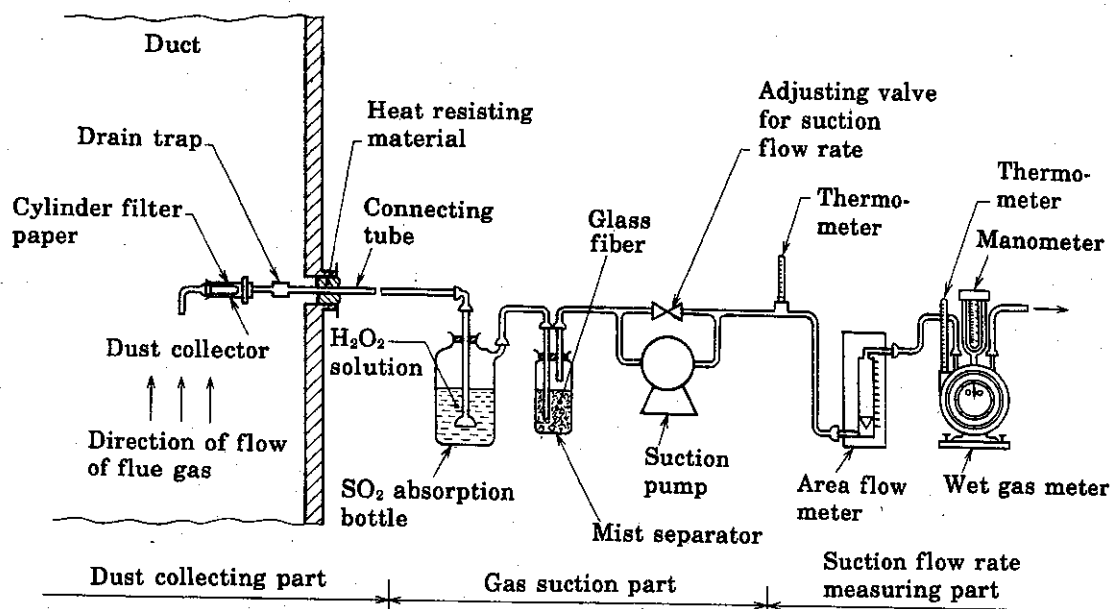
8.3 Ordinary-type sampling apparatus

8.3.1 Ordinary-type manual sampling apparatus The ordinary-type manual sampling apparatus shall be as follows.

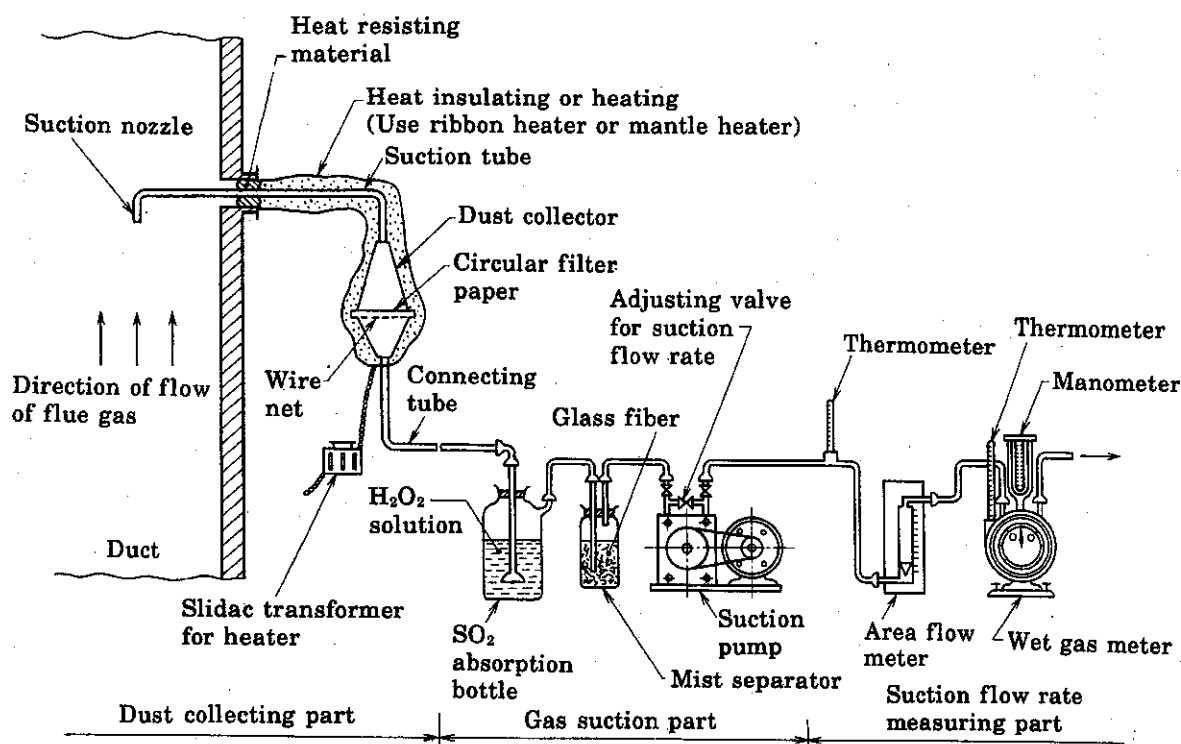
- (1) Arrangement The arrangement of ordinary-type manual sampling apparatus shall be in accordance with the basic arrangement of the sampling apparatus for dust specified in 8.2. The ordinary-type manual sampling apparatus of type 1 is shown in Figure 8 (1), and that of type 2 is shown in

Fig. 8. Example of arrangement of ordinary-type manual sampling apparatus

(1) Type 1

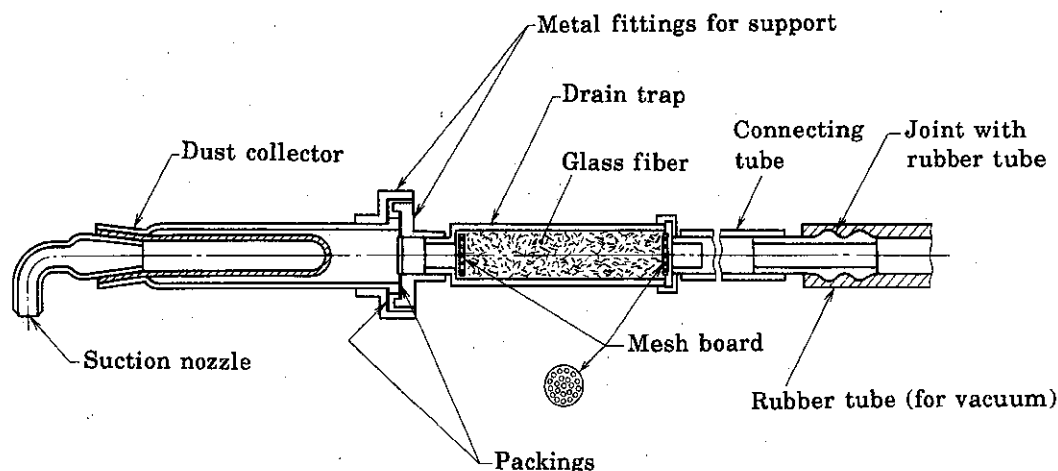


(2) Type 2



- (2) Dust collecting part The dust collecting part shall consist of a suction nozzle, dust collector, metal fittings for support, drain trap, connecting tube, and others as described below. For type 2, however, the suction tube shall be connected behind the suction nozzle. An example of structure of dust collecting part is given in Fig. 9.

Fig. 9. Example of structure of dust collecting part
(Type 1 using cylinder filter paper)



(2.1) Suction nozzle The suction nozzle shall meet the following requirements:

- (a) It shall not disturb the gas flow at both inside and outside of the suction nozzle, and shall have the inside diameter d of 4 mm or more.
- (b) Inside diameter d of suction nozzle shall be measured accurately to the nearest 0.1 mm.
- (c) The tip of the suction nozzle shall be finished to an acute angle of 30° or less as shown in Fig. 10 or shall have a smooth hemispherical form.
- (d) The suction nozzle shall have smooth surfaces at both inside and outside.
- (e) The suction tube from the suction nozzle to the dust collector shall have the smooth inside surface without any abrupt change in its section and bend.
- (f) The material of a suction nozzle shall be hard glass, silica, stainless steel, or the material having properties equivalent or superior to them. The example of the shape of suction nozzle is given in Fig. 11.

Fig. 10. An example of tip
of suction nozzle

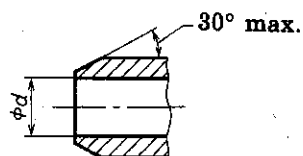
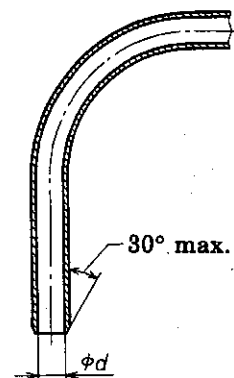


Fig. 11. An example of the shape
of suction nozzle



(2.2) Dust collector

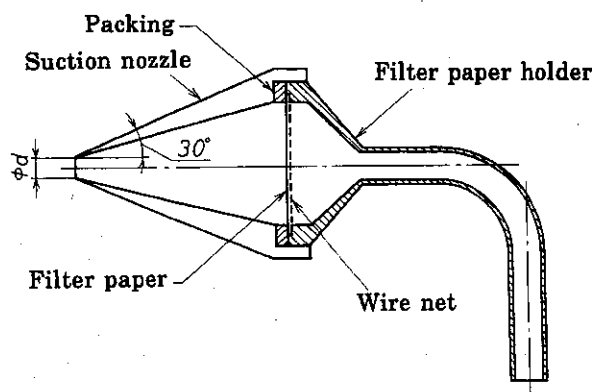
- (a) The dust collector to be used shall filtrate sampling of dust by filter paper, and its performance shall be 99 % or more in the dust collection efficiency. It shall not cause chemical change during the use.
- (b) The dust collector using filter paper shall consist of a suction nozzle, filter paper, and a holder of filter paper.
- (c) There are such shapes as circular, cylinder, and so on among the filter paper, and it shall be attached to the suitable filter paper holder according to its shape. However, the filter paper having 30 mm or larger in effective diameter shall be used for circular-type.

An example of the dust collector using the circular filter paper is shown in Fig. 12, and an example of the dust collector using the cylinder filter paper, in Fig. 13.

- (d) Use the filter paper holder suitable to the shape of filter paper, of which structure is capable of easily attaching or detaching filter paper. The same material as that specified in (f) of (2.1) shall be used.
- (e) No leakage of air shall exist at the place where the filter paper is attached.
- (f) In the case of coexistence of mist, either of the following methods shall be used.
 - ① By using high strength filter paper, or by fitting corrosion resistant wire net or the like behind filter paper as shown in Fig.14, the filter paper is made not to be broken, and besides a mist separator is used at the latter stage. An example of the construction of a sampling apparatus in this case is given in Fig. 15.
 - ② A dust collector with an impactor given in Fig. 16 is used to separate the mist preliminarily.

Fig. 12. Examples of dust collector using circular filter paper

(1) Type 1



(2) Type 2

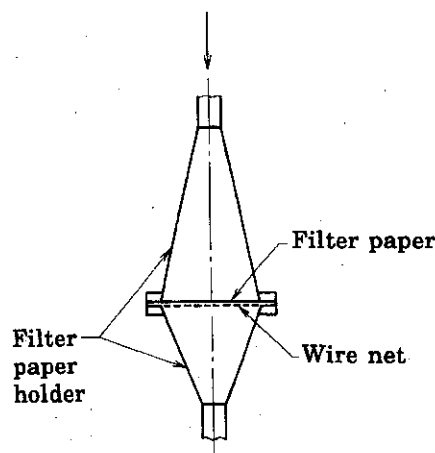
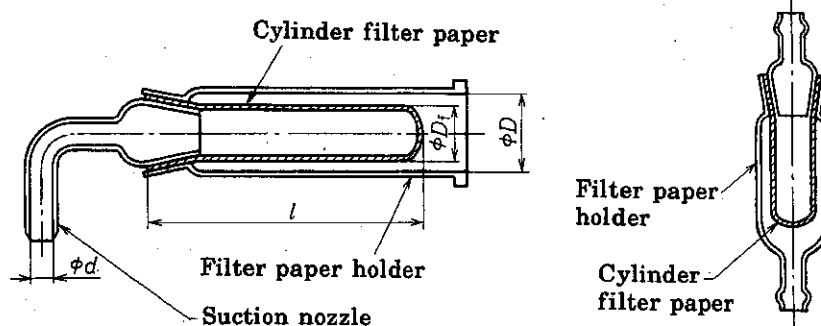


Fig. 13. Examples of shape and dimensions of dust collector using cylinder filter paper

(1) Type 1

(2) Type 2



Unit: mm

D	24	30	35	45
d	4, 6, 8, 10, 12, 14, 16, 18, 20, ...			
D_f	20	25	30	40
l	90			

Fig. 14. Example of dust collector using wire net-supported filter paper

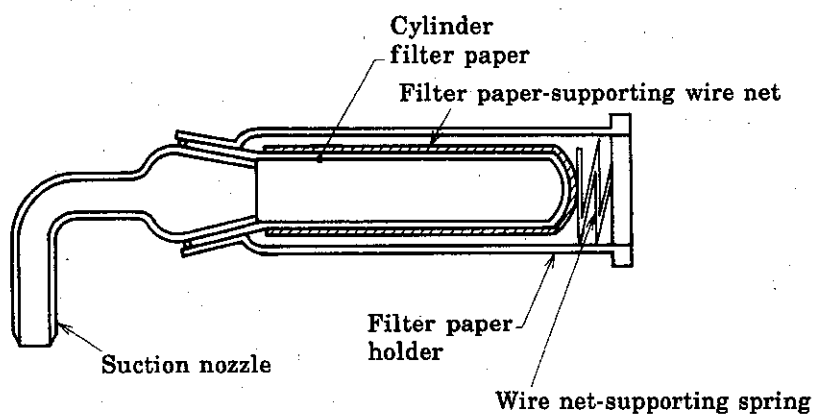


Fig. 15. Example of arrangement of sampling apparatus for particle accompanying mist

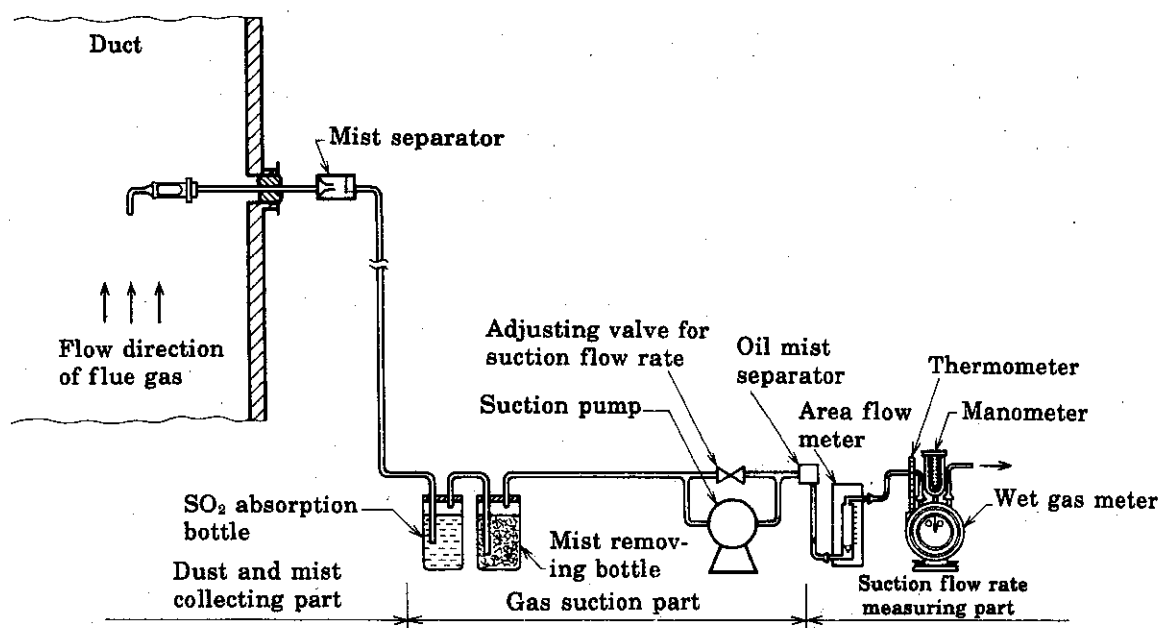
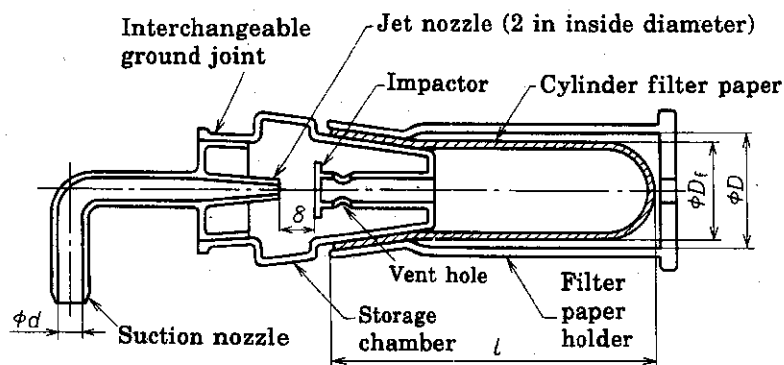


Fig. 16. Example of shape and dimensions of dust collector equipped with an impactor

Unit: mm



Remarks: The dimensions shall follow the example of Fig. 13.

- (g) The dust collector shall be selected taking into consideration the performance of filtration media shown in Table 3 at the time of use.

Remarks: Some of filtration media has the properties to reduce its mass by heating, and to adsorb or react with the coexisting material (such as sulfur oxide) in flue gas, therefore sufficient care on these properties shall also be taken for the selection of filter since they may bring about a measuring error.

Table 3. Performance of filtration media of dust collectors

Item	Dust collector using filter paper			
	Glass fiber	Silica fiber	Fluororesin	Membrane
Temperature at the time of use	500°C max.	1000°C max.	250°C max.	110°C max.
Collecting efficiency	99% or over			
Pressure loss	Under 1.96 kPa		Under 5.88 kPa	
Hygroscopicity	Under 1%		Under 0.1%	Under 1%

Remarks: The values in this table are based on the testing method of 5. in JIS K 0901.

(2.3) Support metal fitting, drain trap, and connecting tube The support metal fitting, drain trap and the connecting tube shall be as follows: (Refer to Fig. 9). The material shall be stainless steel bar specified in JIS G 4303, or that equivalent or superior thereto.

- (a) Support metal fitting For holding the dust collector.
- (b) Drain trap With glass fiber or silica fiber packed its inside, and for preventing the drain owing to suction gas from reverse flow into the dust collector.
- (c) Connecting tube This is the connection between the dust collector or the drain trap and the rubber tube (for vacuum) for sucking gas.

Remarks: Glass tube may be used for connection.

(3) Gas suction part Use the apparatus specified in 6.1.2 (2).

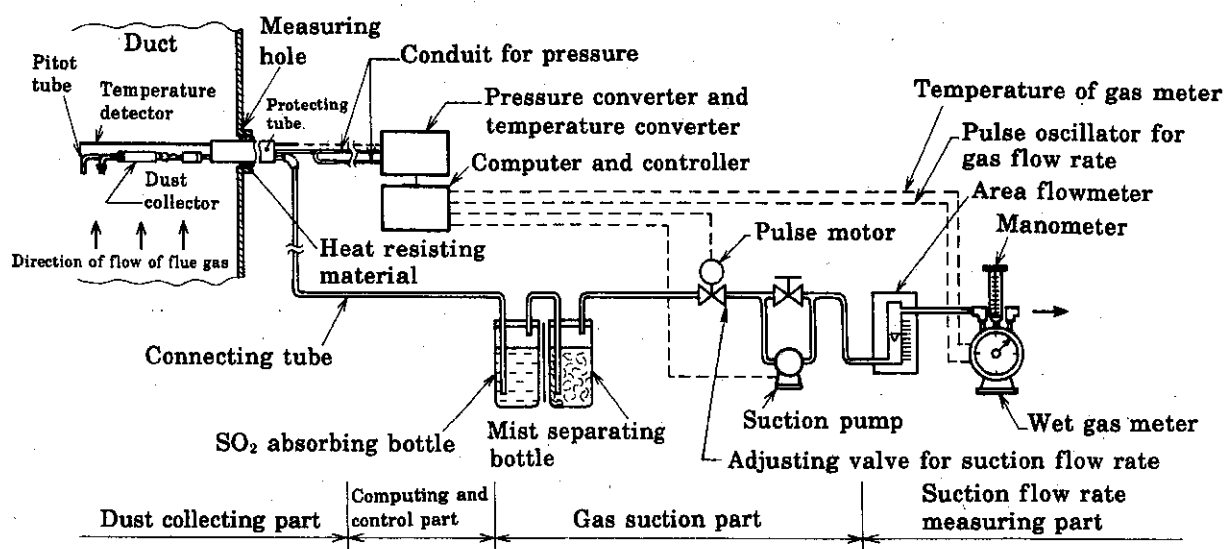
(4) Measuring part for suction flow rate Use the apparatus specified in 6.1.2 (3).

8.3.2 Ordinary-type automatic sampling apparatus The ordinary-type automatic sampling apparatus shall be as follows.

(1) Arrangement For the arrangement of ordinary-type automatic sampling apparatus, operation and control part is added to the basic arrangement specified in 8.2.

An example of the arrangement of ordinary-type automatic sampling apparatus is given in Fig. 17.

Fig. 17. Example of the arrangement of ordinary-type automatic sampling apparatus



- (2) **Dust collecting part** The dust collecting part is composed of a suction nozzle, dust collector, Pitot tube, temperature detector, pressure conduit, and protective tube.
- Suction nozzle and dust collector** The suction nozzle and the dust collector specified in 8.3.1 (2) (2.1) and (2.2) are used.
 - Pitot tube** The Pitot tube specified in 7.2 (1) is used, which is set near the suction nozzle.
 - Temperature detector** The temperature detector specified in 5.2 is used, which is set near the suction nozzle.
 - Pressure conduit** The pressure conduit transmits pressure obtained from Pitot tube to pressure converter outside the duct. The material of the part inserted into the duct shall be stainless steel specified in JIS G 3448 or heat-resistance rubber or the like.

Further, for the part outside the duct, a rubber tube or the like is used.

- Protective tube** The protective tube protects the dust collecting part, which is so constructed that flue gas does not leak through the measuring hole.

Remarks: The protective tube can be omitted when not required.

- (3) **Computing and control part** The computing and control part is composed of a pressure converter, temperature converter, computer and controller as follows.
- Pressure converter** The pressure converter so constructed that it converts dynamic pressure of a measuring point, its static pressure (total pressure), and atmospheric pressure in measurement to an electric signal to be inputted to computer is used.
 - Temperature converter** The temperature converter so constructed that temperature of flue gas at a measuring point, and temperature⁽³⁾ of a gas meter in measurement are converted to an electric signal, and inputted to computer, is used.

Note (3) In the case of using a wet gas meter, saturated water vapor corresponding to temperature of suction gas is automatically inputted to computer.

- (c) Computer The computer calculates flow velocity of flue gas according to the formula (7) specified in 7.3.4 by using the dynamic pressure, static pressure (total pressure), flue gas temperature at the measuring point converted electrically with the pressure converter and the temperature converter, and atmospheric pressure in measurement, and further the Pitot tube coefficient to be inputted and density of flue gas in the duct.

Further, the computer calculates simultaneously isokinetic sampling flow rate at the measuring point according to the formula (11) specified in 9.3.1 by using the calculated velocity, the inputted inner diameter of a suction nozzle, the moisture content in the flue gas obtained preliminarily according to the specification of 6., and the temperature of a gas meter converted electrically with the temperature converter.

The computer so constructed that a control signal is sent to a control motor so that suction flow rate as a pulse signal from the gas meter⁽⁴⁾ equipped with a pulse oscillator and a suction gas temperature detector is equalized to the isokinetic sampling flow rate, is used.

Note (4) The gas meter oscillating at least one pulse per 0.01 l suction gas amount.

Remarks: The computing and control part can be equipped with attached instruments which indicate and record various measured values or calculated values as well as dynamic pressure, static pressure, flue gas temperature, and the state of both isokinetic sampling flow rates.

- (d) Controller The controller so constructed that controls an automatic adjusting valve so that the isokinetic sampling flow rate calculated with the computer is equalized to the suction flow rate to be sucked, is used.

- (4) Gas suction part The gas suction part specified in 6.1.2. (2) is used. However, the adjusting valve for suction flow rate so constructed that suction flow rate is controlled so that the isokinetic sampling flow rate calculated with computer is equalized to the gas flow rate to be sucked, and the adjusting valve is opened and closed by driving the attached control motor with a control signal from the controller to be capable of adjusting the suction flow rate, is used. The setting shall be made on a suitable position in the flow rate controller or of the gas suction part.

- (5) Suction flow rate measuring part The suction flow rate measuring part specified in 6.1.2 (3) is used. As the integrating flowmeter, a wet or dry gas meter equipped with a pulse oscillator and a suction gas temperature detector is used.

In the case of using the dry gas meter, gas shall be dried prior to introducing into the meter.

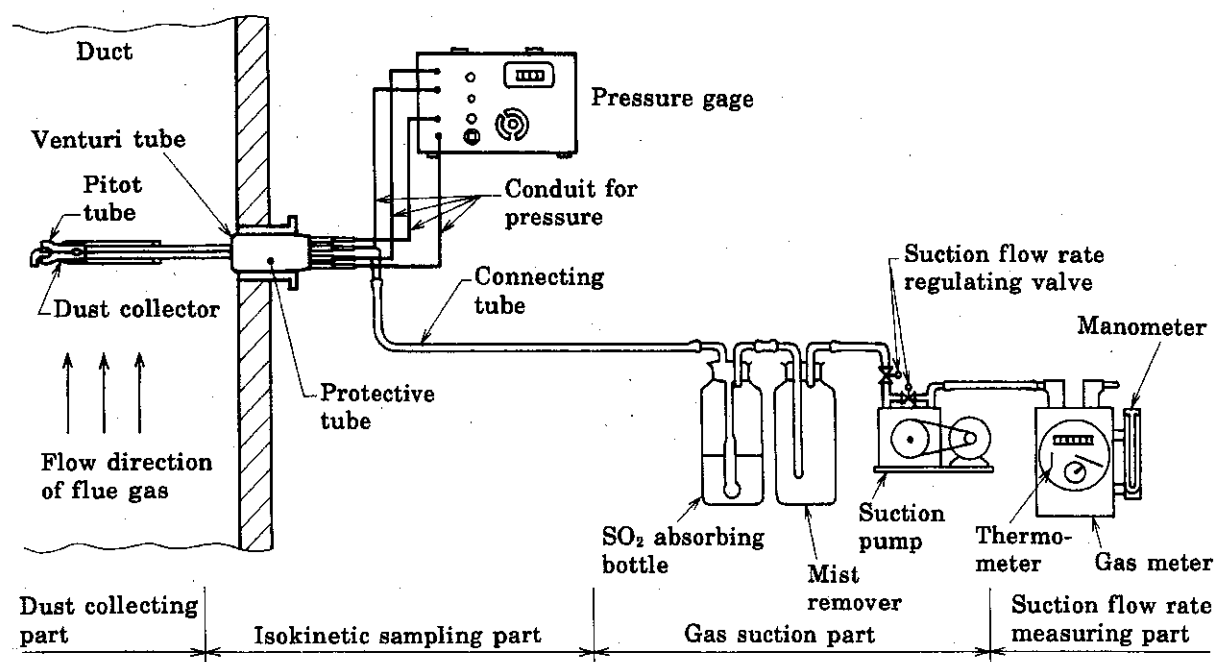
8.4 Performance of ordinary-type automatic sampling apparatus and its performance testing method The performance of an ordinary-type automatic sampling apparatus and its performance testing method shall be in accordance with Annex.

8.5 Balanced-type sampling apparatus

8.5.1 Balanced-type manual sampling apparatus The balanced-type manual sampling apparatus shall be as follows.

- (1) **Arrangement** In a balanced-type manual sampling apparatus, an isokinetic sampling part is added to the basic arrangement of a dust sampling apparatus specified in 8.2. An example of the arrangement of the balanced-type manual sampling apparatus is given in Fig. 18.
- (2) **Dust collecting part** The dust collecting part specified in 8.3.1 (2) is used.
- (3) **Isokinetic sampling part** The isokinetic sampling part shall be composed of an isokinetic sampling mechanism, conduit for pressure, protective tube, and pressure gage as follows.
 - (a) **Isokinetic sampling mechanism** The isokinetic sampling mechanism is a system of using dynamic pressure of flue gas and has functions wherein dynamic pressure of flue gas at a measuring point is measured with the Pitot tube installed near a suction nozzle, besides a differential pressure of the gas sucked is measured with a Venturi tube connected behind the dust collector, and isokinetic sampling is obtained by regulating the flow rate of suction gas so that the differential pressure is equalized to the dynamic pressure, which has a construction capable of connecting the dynamic pressure and the differential pressure to the conduit for pressure.
 - (b) **Conduit for pressure** The conduit for pressure specified in (d) of 8.3.2 (2) is used.
 - (c) **Protective tube** The protective tube specified in (e) of 8.3.2 (2) is used.
 - (d) **Pressure gage** The pressure gage specified in 7.2 (2) is used, which introduces the dynamic pressure and differential pressure transmitted through the conduit for pressure from an isokinetic sampling mechanism alternately with a change-over valve, and indicates that.
 Remarks: Dynamic pressure and differential pressure may be indicated simultaneously with two pressure gages.
- (4) **Gas suction part** The gas suction part specified in 6.1.2 (2) is used.
- (5) **Suction flow rate measuring part** In the suction flow rate measuring part, the wet or dry gas meter specified in 6.1.2 (3) is used. However, an instantaneous flowmeter may be omitted.

Fig. 18. Example of arrangement of balanced-type manual sampling apparatus (dynamic pressure type)



8.5.2 Balanced-type automatic sampling apparatus The balanced-type automatic sampling apparatus shall be as follows.

- (1) Arrangement In the balanced-type automatic sampling apparatus, an isokinetic sampling part is added to the basic arrangement of the dust sampling apparatus specified in 8.2. An example of the arrangement of the balanced-type automatic sampling apparatus is given in Fig. 19.
- (2) Dust collecting part The dust collecting part specified in 8.5.1 (2) is used.
- (3) Isokinetic sampling part The isokinetic sampling part is composed of an isokinetic sampling mechanism, conduit for pressure, protective tube, and computing control unit as follows.
 - (a) Isokinetic sampling mechanism The isokinetic sampling mechanism has a function where if the flow rate of suction gas is regulated so that the dynamic pressure (differential pressure) or static pressure of the gas to be sucked is equalized to the dynamic pressure or static pressure of flue gas at a measuring point, isokinetic sampling can be carried out, which is classified into dynamic pressure type and static pressure type according to the pressure to be used. An example of its structure and arrangement are given in Fig. 20.

The isokinetic sampling mechanism specified in (a) of 8.5.1 (3) is used as the dynamic pressure type one.

The static pressure type isokinetic sampling mechanism has a function where if the flow rate of suction gas is adjusted so that the static pressure of suction gas measured with the static pressure hole provided in the suction nozzle is equalized to the static pressure at the measuring point measured with the Pitot tube installed near the suction nozzle, isokinetic sampling can be carried out, which has a

structure capable of transmitting the dynamic pressure and differential pressure to the computing control unit through the conduit for pressure.

- (b) Conduit for pressure The conduit for pressure specified in (d) of 8.3.2 (2) is used.
- (c) Protective tube The protective tube specified in (e) of 8.3.2 (2) is used.
- (d) Computing control unit The computing control unit, equipped with the pressure converter and temperature converter which convert both kinds of pressure and flue gas temperature transmitted from the isokinetic sampling mechanism to electric signals and indicate them, having a function wherein the flow rate of suction gas is adjusted by opening and closing the suction flow rate regulating valve by driving the motor connected thereto by a control signal so that both kinds of pressure are equalized if a difference exists by comparing them with each other is used.

Remarks: Attached instruments which indicate and record various measured values or calculated values or the like as well as both kinds of pressure, flue gas temperature, and the equilibrium condition of both kinds of pressure, can be provided to the computing control unit.

- (4) Gas suction part The gas suction part specified in 8.3.2 (4) is used.
- (5) Suction flow rate measuring part As the suction flow rate measuring part, that specified in 8.3.2 (5), or a gas meter equipped with a pulse oscillator and a suction gas temperature detector is used.

Fig. 19. Example of arrangement of balanced-type automatic sampling apparatus

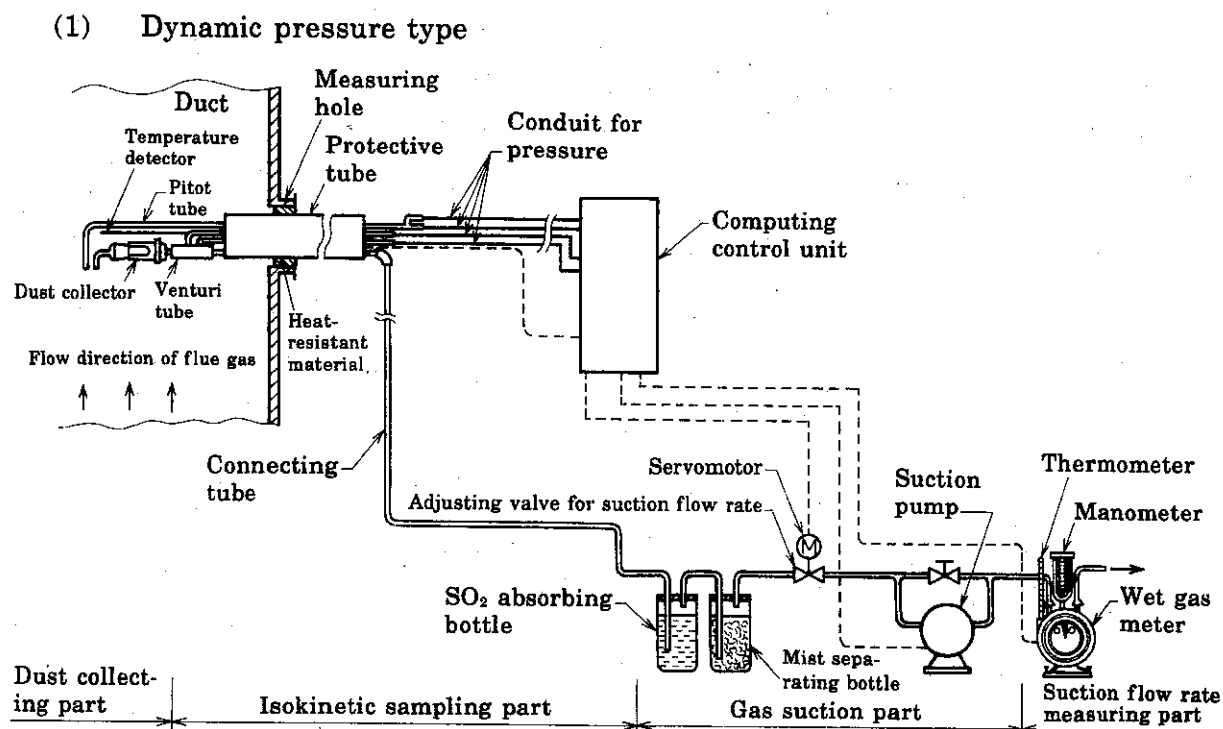


Fig. 19. (continued)

(2) Static pressure type

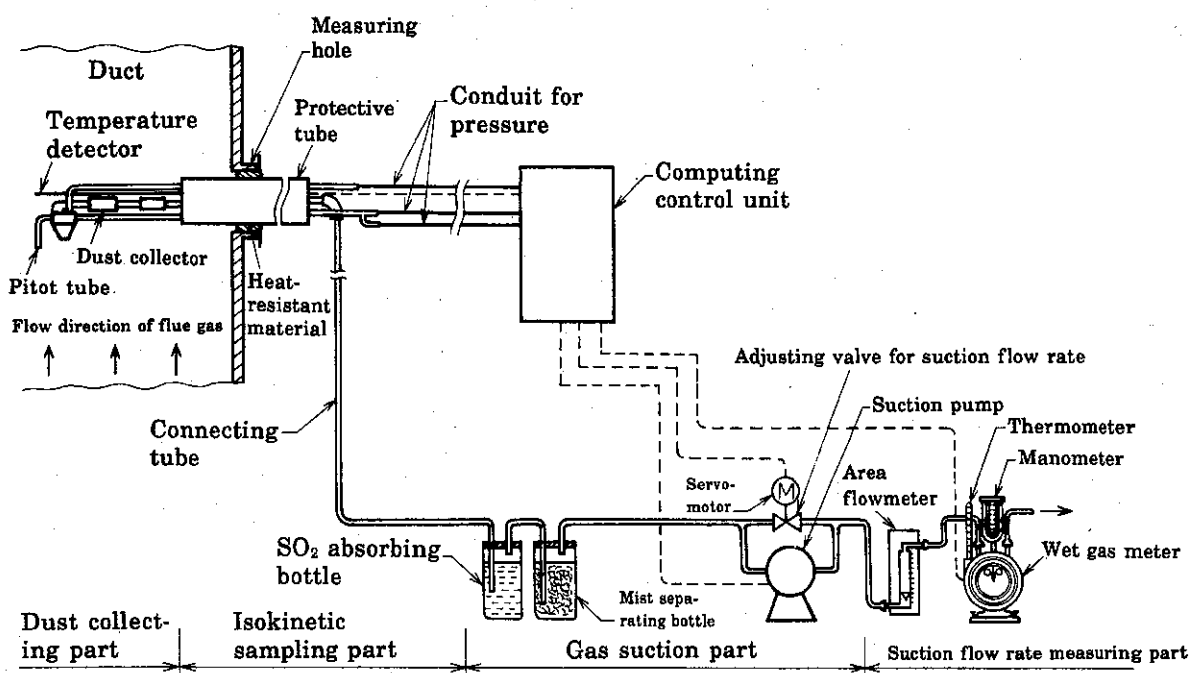
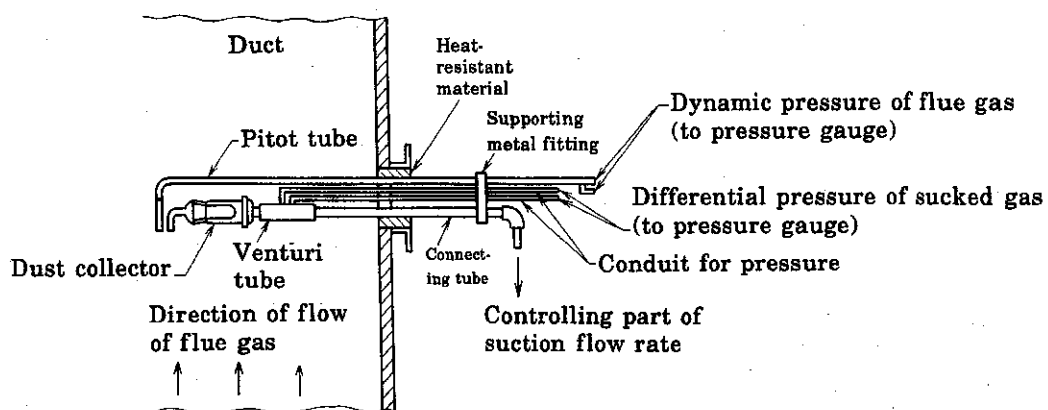


Fig. 20. Example of structure and arrangement of isokinetic sampling mechanism

(1) Dynamic pressure type



Structure of venturi tube

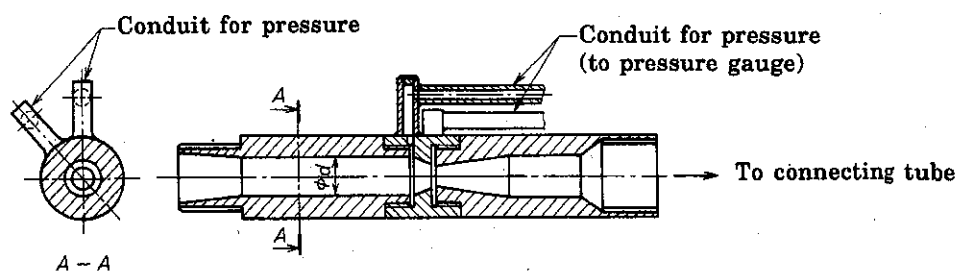
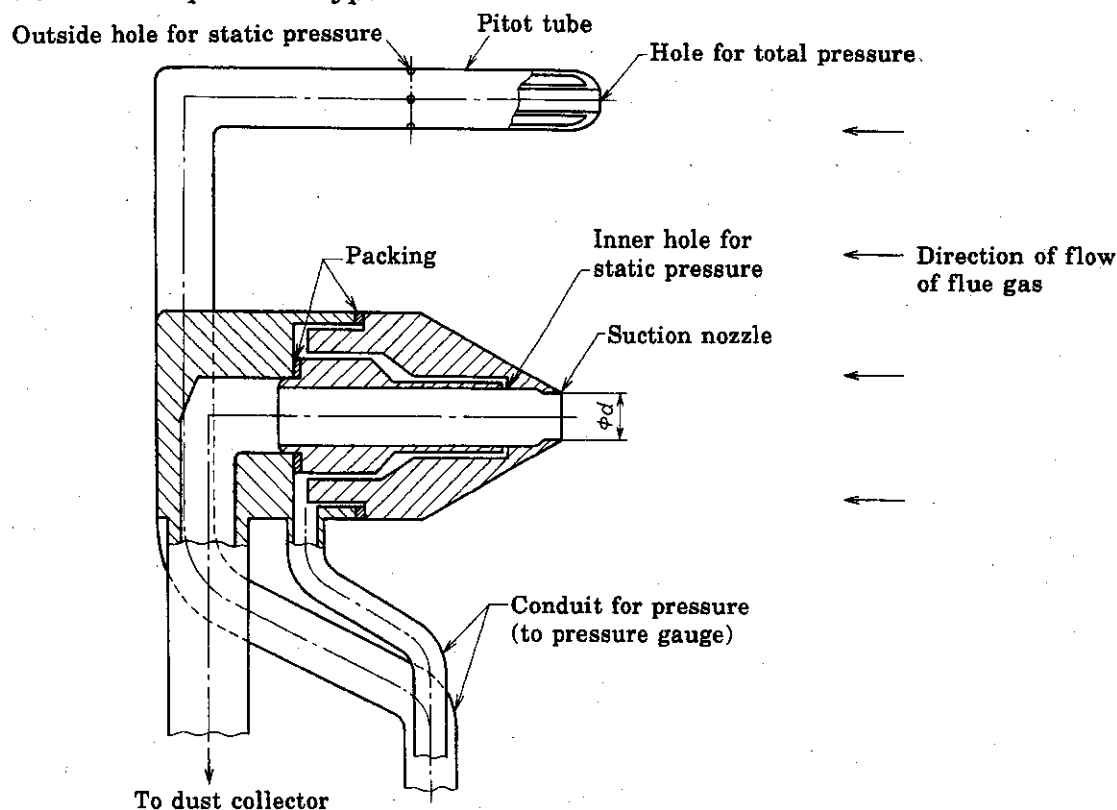


Fig. 20. (continued)

(2) Static pressure type



8.6 Performance and its testing method of balanced-type sampling apparatus
The performance and its testing method of balanced-type sampling apparatus shall be in accordance with Annex.

9. Sampling method for dust

9.1 Measuring points Select the measuring points of sampling for dust according to 4.3.

9.2 Kind of sampling method for dust The sampling method for dust shall be in accordance with anyone of the following.

- (1) Method of each point sampling Take a dust sample by using one dust collector at each measuring point.
- (2) Method of traverse sampling Traverse the measuring points by using one dust collector to take a dust sample during the same sucking time at each point.
- (3) Method of representative point sampling Take a dust sample by using one or several dust collectors at representative points in accordance with the description in Remarks 3. of 4.3.

9.3 Preparation of dust sampling

9.3.1 Calculation of suction flow rate for isokinetic sampling When the ordinary-type sampling apparatus is used, the suction flow rate for an isokinetic sampling shall be obtained from the following formula (11).

$$q_m = \frac{\pi}{4} d^2 v \left(1 - \frac{x_w}{100} \right) \frac{273 + \theta_m}{273 + \theta_s} \times \frac{P_a + P_s}{P_a + P_m + P_v} \times 60 \times 10^{-3} \dots\dots (11)$$

- where, q_m : flow rate in isokinetic sampling at gas meter (l/min)
- d : inner diameter of suction nozzle (mm)⁽⁵⁾
- v : flow velocity of flue gas (m/s)⁽⁶⁾
- x_w : volume percentage of water vapor in flue gas (%)⁽⁶⁾
- θ_m : temperature of suction gas at gas meter (°C)⁽⁶⁾
- θ_s : temperature of flue gas (°C)⁽⁶⁾
- P_a : atmospheric pressure (kPa){mmHg}⁽⁶⁾
- P_s : static pressure at measuring point (kPa){mmHg}⁽⁶⁾
- P_m : gauge pressure of suction gas at gas meter (kPa){mmHg}⁽⁶⁾
- P_v : saturated water vapor pressure at θ_m (kPa){mmHg}

When using a dry gas meter and having dried the gas prior to introducing into the meter, it is calculated excepting the term P_v in formula (11).

Notes ⁽⁵⁾ The selection of d shall be according to 9.3.2 (1).

⁽⁶⁾ For the value of θ_s , the value measured according to the specification of 5., for the values of x_w , θ_m , P_m , and P_v , the values measured according to the specification of 6. and for the values of v , P_a , and P_s , the values measured according to the specification of 7. may be used.

9.3.2 Preparation of dust collector The preparation of dust collector shall be carried out as follows:

- (1) Choose the inner diameter of suction nozzle and the dimensions of a filter paper referring the calculation in 9.3.1 in order to make the apparent flow velocity of the gas passing through the filter paper, generally, not more than 0.5 m/s.
- (2) Dry satisfactorily the filter paper at 105°C to 110°C, and after cooling in a desiccator to room temperature, weigh the mass. When the temperature of flue gas is 100°C or more, however, heat beforehand the filter paper to get constant weight at the approximately same temperature as that of flue gas, and weigh it similarly to the above.

Use the weighing balance having the reciprocal sensibility of 0.1 mg or below.

Remarks: Weighing should preferably be carried out under the environment of about 50 % in relative humidity.

Cares shall be sufficiently taken for influences owing to charging in weighing.

- (3) Keep the filter paper that has been weighed, in a dust-free vessel and the like until the time of use to protect them from depositing dust.

9.4 Sampling of dust The sampling of dust shall be carried out as follows:

- (1) Use the sampling apparatus for dust specified in 8, and after attaching the prepared dust collector by using supporting metal, confirm the air-tightness of the whole apparatus.
- (2) Insert the suction nozzle to the measuring point in the direction against the flow of flue gas, and instantly beginning the suction, face the nozzle rightly to the flow of flue gas, followed by sucking in with the method of the isokinetic sampling. The permissible range shall be as follows:
 - (a) The deviation of flowing direction of flue gas and the direction of a suction nozzle at a measuring point shall be not more than 10°.
 - (b) The relative error of the flow velocity of the gas sucked from the suction nozzle to the flow velocity of flue gas at a measuring point shall be within the range of -5 % to +10 %.
 - (c) When using an ordinary-type manual sampling apparatus, carry out the isokinetic sampling after adjusting quickly to the suction flow rate obtained in advance according to 9.3.1.
 - (d) When using a balanced-type manual sampling apparatus, after adjusting quickly the flow rate of suction gas to equalize the dynamic pressure of flue gas to differential pressure of suction gas, carry out the isokinetic sampling.
- (3) Select the volume of sucked gas so that the amount of dust collected becomes as follows:
 - (a) In the case of circular filter paper, it shall be about 0.5 mg per 1 cm² of collecting area.
 - (b) In the case of cylinder filter paper, total collected amount shall be not less than 5 mg.

Remarks: When applying (a) and (b), be careful specially about the managing, handling, and weighing of filter papers.

- (4) Stop suction after finishing sucking of flue gas, and measure the volume of sucked gas. After making again the nozzle's direction against to the flow of flue gas, take it out promptly from the duct.

Until this time, never disconnect the piping of downstream from the dust collector. In the case of a large negative pressure in the duct, however, measure the volume of sucked gas at the time of making the nozzle against the flow, and take out the nozzle quickly from the duct while the suction pump is operating. Then stop the pump.

- (5) Dry the filter paper which has collected dust, generally for 1 h at 105°C to 110°C, and after cooling in a desiccator to room temperature, weigh the mass.
- (6) In the case of coexistence of mist in flue gas, use the system specified in 8.3.1 (2) (2.2) (f), besides in the case of dust contained in the separated mist, or if there is a possibility of containing soluble substance (particle),

recover the mist, sufficiently dry, as a rule, at 105°C to 110°C, cool it in a desiccator to room temperature and thereafter, weigh and add that to the collected dust.

- (7) If there is a possibility that the flue gas may contain such as sulfuric acid mist which may give influence to measuring values, use the filter paper⁽⁸⁾ having no reactivity with these substances, heat the filter paper which has collected the sample for about 2 h at about 250°C, and weigh its mass after cooling in a desiccator to room temperature.

Note ⁽⁸⁾ The filter paper made of fluororesin or of silica fiber treated with sulfuric acid may be used for this purpose.

Remarks: Alternative treatment may be adopted as follows: extract the filter paper deposited with dust with warm water, determine the sulfuric acid by chemical analysis, and subtract the quantity equivalent to sulfuric acid from the mass of dust.

- (8) In the case of type 2 of ordinary-type sampling apparatus, add the dust adhered inside of suction pipe from the nozzle to the filter paper, to the amount of dust collected already.
- (9) Obtain the mass of dust collected from the mass difference of the filter paper or the dust collector before and after collecting.

9.5 Measuring method for volume of suction gas The measuring of volume of suction gas specified in 9.4 (4) shall be made with the integrating flow meter or instantaneous flow meter specified in 6.1.2 (3). Measuring methods shall be as follows:

- (1) For ascertaining the suction time, record the times of the commencement and the finishing of the suction.
- (2) Read the graduations of gas meter to the nearest 0.1 l at the times of the commencement and the finishing of the suction.
- (3) During the suction, measure the temperature and pressure of the suction gas in the gas meter.
- (4) Obtain the volume of sucked dry gas under the standard condition from the following formula (12):

$$V'_N = V_m \times \frac{273}{273 + \theta_m} \times \frac{P_a + P_m - P_v}{101.3} \times 10^{-3} \dots\dots\dots (12)$$

where, V'_N : volume of sucked dry gas under standard condition (m^3_N)

V_m : volume of sucked gas (reading of wet gas meter) (l)

θ_m : temperature of sucked gas at gas meter (°C)

P_a : atmospheric pressure (kPa) {mmHg}

P_m : gauge pressure of gas at gas meter (kPa){mmHg}

P_v : saturated water vapor pressure at θ_m (kPa){mmHg}

Informative reference: When using mmHg as the unit of pressure, adopt 760 instead of 101.3 in the formula (12).

When using a dry gas meter and having dried the gas prior to introducing to the meter, the volume shall be calculated, excepting the term P_v in the formula (12) and by reading V_m as the volume of sucked dry gas (reading at dry gas meter).

10. Calculation of dust concentration

10.1 Dust concentration in dry flue gas at each measuring point The dust concentration in flue gas shall be expressed by the mass of dust contained in 1 m³ of dried flue gas which has been converted into standard condition (0°C in temperature, 101.3 kPa {760 mmHg} in atmospheric pressure), and it can be obtained according to the following formula (13).

$$C_N = \frac{m_d}{V'_N} \dots\dots\dots (13)$$

where, C_N : dust concentration in dry flue gas (g/m³_N)
 m_d : mass of collected dust (g)
 V'_N : volume of sucked dry flue gas under the standard condition (m³_N)

10.2 Average dust concentration in dry flue gas on whole section The average dust concentration can be obtained by making use of dust concentration at each divided section, according to the following formula (14):

$$\bar{C}_N = \frac{C_{N1} \cdot A_1 \cdot v_1 + C_{N2} \cdot A_2 \cdot v_2 + \dots + C_{Nn} \cdot A_n \cdot v_n}{A_1 \cdot v_1 + A_2 \cdot v_2 + \dots + A_n \cdot v_n} \dots\dots\dots (14)$$

where, $\bar{C}_N$: average dust concentration on whole section (g/m³_N)
 $C_{N1}, C_{N2}, \dots, C_{Nn}$: dust concentration at each section (g/m³_N)
 $A_1, A_2, \dots, A_n$: area of each section (m²)
 $v_1, v_2, \dots, v_n$: flow velocity of gas at each section (m/s)

Remarks: When the dust concentration is measured by the method of traverse sampling, the average dust concentration of whole section shall be calculated in accordance with the above-mentioned method.

10.3 Rounding off of measured values of dust concentration The measured values of dust concentration shall be rounded off to two significant figures in accordance with JIS Z 8401.

Informative reference: Dust concentration in wet flue gas When the dust concentration in the flue gas is expressed by the mass of the dust contained in 1 m³ of wet flue gas in the gaseous state in the duct, it can be obtained according to the following formula (15).

$$C = \frac{m_d}{V_s} \dots\dots\dots (15)$$

$$V_s = V_m \times \frac{100}{100 - x_w} \times \frac{273 + \theta_s}{273 + \theta_m} \times \frac{P_a + P_m - P_v}{P_a + P_s} \times 10^{-3}$$

- where, C : dust concentration in wet flue gas in gaseous state in duct (g/m^3)
- m_d : mass of dust collected (g)
- V_s : volume of sucked gas which has been converted into the gaseous state in a duct (m^3)
- V_m : volume of sucked gas (reading of wet gas meter) (l)
- x_w : volume percentage of water vapor in flue gas (%)
- θ_s : temperature of flue gas ($^{\circ}\text{C}$)
- θ_m : temperature of sucked gas at wet gas meter ($^{\circ}\text{C}$)
- P_a : atmospheric pressure (kPa){mmHg}
- P_s : static pressure at measuring point (kPa){mmHg}
- P_m : gauge pressure of gas at wet gas meter (kPa){mmHg}
- P_v : saturated water vapor pressure at temperature θ_m (kPa){mmHg}

11. Calculation of flow rate of dust The flow rate of dust in the flue gas shall be expressed by the mass of dust contained in flow rate of dry or wet flue gas which has been converted into the standard condition [temperature 0°C , atmospheric pressure 101.3 kPa {760 mmHg}], and it can be obtained according to the following formula (16):

$$S = \bar{C}_N Q'_N \times 10^{-3} = \bar{C}_N Q_N \left(1 - \frac{x_w}{100}\right) \times 10^{-3} \dots\dots\dots (16)$$

- where, S : flow rate of dust in flue gas (kg/h)
- $\bar{C}_N$: average dust concentration for whole section (g/m^3_N)
- Q_N : flow rate of wet flue gas (m^3_N/h)
- Q'_N : flow rate of dry flue gas (m^3_N/h)
- x_w : volume percentage of water vapor in flue gas (%)

12. Record of measured values The result obtained by the measurements and calculations in accordance with this Standard shall be summarized and recorded on the items generally given in Table 4 according to the type of dust sampling apparatus.

Table 4. Recording items of measured values

Measuring items		Ordinary-type sampling apparatus	Balanced-type sampling apparatus	Remarks
General items	1 Date of measurement		○	
	2 Place of measurement		○	
	3 Conditions of generation source			
	(1) Kind of generation source		○	
	(2) Kind of fuel or heat source used		○	
	(3) Position of measurement		○	Show by a diagram.
Items concerning measurement of dust concentration	(4) Sectional shape and approximate dimensions of duct		○	Show by a diagram.
	(5) Number and positions of measuring points		○	Show by a diagram.
	4 Conditions for sampling of dust			
	(1) Type of sampling apparatus		○	Describe clearly either of type 1 or type 2.
	(2) Kind of sampling method		○	
	(3) Inner diameter of suction nozzle		○	
	(4) Material, shape, and dimensions of filter		○	
	5 Items on measurement and calculation			
	(1) Atmospheric pressure	○	—	
	(2) Temperature of flue gas	○	—	
	(3) Moisture content of flue gas	○	—	
	(4) Density of flue gas	○	—	
	(5) Dynamic pressure of flue gas	○	—	
	(6) Static pressure of flue gas	○	—	
	(7) Flow velocity of flue gas	○	—	
	(8) Isokinetic sampling flow rate	○	—	
	(9) Temperature of sucked gas at gas meter	○	○	
	(10) Saturated water vapor pressure at the above mentioned temperature	○	○	Not required for dry gas meter
	(11) Gage pressure at gas meter	○	○	
	(12) Suction gas volume	○	○	
	(13) Time for suction	○	○	
	(14) Mass of collected dust	○	○	
Others	6 Dust concentration	○	○	
	7 Measuring items other than dust concentration			
	(1) Volume of wet flue gas		○	Items on and after this line are calculated, if necessary.
	(2) Volume of dry flue gas		○	
	(3) Dust flow rate		○	

Remarks: The ○ marked items in the table are to be recorded, and the — marked items may be omitted from record.

Attached Table 1. Cited standards

JIS B 7306	Hygrographs
JIS B 7307	Barographs
JIS B 7413	Etched-stem mercury-in-glass thermometers (partial immersion type)
JIS B 7551	Variable area flowmeters
JIS B 8330	Testing methods for turbo-fans and blowers
JIS C 1601	Indicating thermoelectric thermometers
JIS C 1603	Indicating resistance thermometers
JIS G 3448	Light gauge stainless steel pipes for ordinary piping
JIS G 4303	Stainless steel bars
JIS K 0901	Form, size and performance testing methods of filtration media for collecting airborne particulate matters
JIS K 2301	Fuel gases and natural gas – Methods for chemical analysis and testing
JIS K 8125	Calcium chloride (for U-tube)
JIS K 8230	Hydrogen peroxide
JIS M 8813	Coal and coke – Methods for ultimate analysis
JIS R 3503	Glass apparatus for chemical analysis
JIS T 8202	Portable type thermal anemometer
JIS Z 8103	Glossary of terms used in instrumentation
JIS Z 8401	Rules for rounding off of numerical values
JIS Z 8704	Temperature measurement – Electrical methods
JIS Z 8705	Method of temperature measurement by liquid-in-glass thermometer
JIS Z 8761	Method of measurement by variable area flowmeters
JIS Z 8762	Measurement of fluid flow by means of orifice plates, nozzles and venturi tubes

Annex Performance and performance testing method
 of ordinary-type automatic sampling apparatus
 and balanced-type sampling apparatus

1. Scope This Annex specifies the performance and performance testing method of the ordinary-type automatic sampling apparatus and the balanced-type manual and automatic sampling apparatus (hereafter referred to as "apparatus") which are specified in 8.3.2, 8.5.1, and 8.5.2 in the text of this Standard.

2. Performance of apparatus The relative error for the isokinetic sampling of the apparatus, when carrying out the test in 3.2, shall fall in the range from -5 % to +10 % specified in 9.4 (2) (b) in this Standard.

3. Testing method for performance

3.1 Testing conditions and testing apparatus

3.1.1 Testing conditions Testing conditions shall be as follows:

- (1) Temperature Within allowable limit of temperature and within 5°C in temperature fluctuation width
- (2) Humidity Within 65 ± 20 % in relative humidity
- (3) Fluctuation of atmospheric pressure Within 2 kPa {15 mmHg} in the fluctuation width of atmospheric pressure
- (4) Voltage of power source ± 2 % of rated voltage
- (5) Frequency of power source ± 0.2 Hz of rated frequency
- (6) Time of warming up Set the apparatus to a state wherein dust is sampled, and warm up for 15 min or more after switching on

3.1.2 Testing apparatus The testing apparatus shall be as follows:

(1) Wind tunnel for test

- (a) Flow velocity range to be set: from 0 m/s to 30 m/s
- (b) Distribution of flow velocity: ± 1 % at effective measuring section
- (c) Fluctuation of flow velocity: ± 5 %/h
- (d) Turbulence coefficient: not more than 0.5%

(2) Measuring instrument

- (a) Pitot tube: L-type Pitot tube specified in JIS B 8330
- (b) Pressure gauge: Pressure gauge specified in 7.2 (2) of the text of this Standard, Askania-type micromanometer or the like having at least 1 Pa {0.1 mmH₂O} resolving power.
- (c) Gas meter: Gas meter specified in 6.1.2 (3) in the text of this Standard.

- (d) Other instruments: Thermometer specified in JIS B 7413, JIS C 1601, and JIS C 1603

Hydrograph specified in JIS B 7306

Barograph specified in JIS B 7307

Thermal anemometer specified in JIS T 8202

3.2 Testing method of relative error for isokinetic sampling of the apparatus

3.2.1 General The testing method of relative error for isokinetic sampling of the apparatus shall be in accordance with 3.2.2. If no wind tunnel for test is available, 3.2.3 may be applicable.

3.2.2 Testing method by wind tunnel for test From among the fixed velocities in the vicinity of 5 m/s, 10 m/s, 15 m/s, 20 m/s, and 30 m/s in a wind tunnel for test, establish three arbitrary velocities respectively where the suction is capable according to the inner diameter of suction nozzle of the apparatus. By using the apparatus per each flow velocity, carry out isokinetic sampling, and obtain the suction flow rate from the sucked air amount and suction time which are measured with a gas meter. Besides, simultaneously measure the flow velocity of air and various conditions (temperature, pressure, humidity, etc.) of air necessary for calculation of isokinetic sampling flow rate according to the requirements specified in 7.3 of the text of this Standard by using the Pitot tube.

Obtain the relative error of isokinetic sampling according to the following method of (1) or (2).

(1) Method for comparing flow velocity

Obtain the flow velocity of suction air according to the following formula (1).

$$v_1 = \frac{q_1}{(\pi/4)d^2} \times \frac{1000}{60} \dots\dots\dots (1)$$

where, v_1 : flow velocity of suction air (m/s)
 q_1 : flow rate of suction air (l/min)
 d : inner diameter of suction nozzle (mm)

Obtain the relative error of isokinetic sampling according to the following formula (2).

$$E = \frac{v_1 - v_0}{v_0} \times 100 \dots\dots\dots (2)$$

where, E : relative error of isokinetic sampling (%)
 v_0 : flow velocity of air in wind tunnel for test (m/s)

(2) Method for comparing suction flow rate

Calculate the flow rate of isokinetic sampling to be obtained by calculation according to the formula (11) specified in 9.3.1 in the text of this Standard, and obtain the relative error of isokinetic sampling according to the following formula (3).

$$E = \frac{q_m - q_1}{q_1} \times 100 \dots\dots\dots (3)$$

where, E : relative error of isokinetic sampling (%)
 q_m : flow rate of isokinetic sampling obtained by calculation (l/min).
 q_1 : flow rate of suction air measured with gas meter (l/min)

3.2.3 Testing method without using wind tunnel for test The test without using a wind tunnel for test shall be as follows.

(1) Testing method to be carried out on job site

- (a) Site test conditions The measuring place shall be the straight part of a duct, which is the place wherein the condition (temperature, density, static pressure, flow velocity, etc.) of flue gas is stable at the measuring place and measuring point which are specified in 4. of the text of this Standard.
- (b) Testing method Suck flue gas for specific time by the sampling method for dust specified in 9.4 of the text of this Standard at a selected site, and obtain the suction flow rate from the amount of suction gas measured with a gas meter. Besides, calculate the flow rate of isokinetic sampling according to the formula (11) specified in 9.3.1 of the text of this Standard from the flow velocity of flue gas measured with the Pitot tube jointly set simultaneously, and the measured values of various conditions of the flue gas and suction gas which are necessary for calculation of the flow rate of isokinetic sampling. Obtain the relative error of isokinetic sampling from the formula (3) of 3.2.2 (2).

(2) Testing method carried out indoors

- (a) Ordinary-type automatic sampling apparatus and balanced-type sampling apparatus Obtain the suction flow rate from the amount of suction air which is sucked for specific time through the suction nozzle of an apparatus against the air flow wherein the flow velocity prepared by using a blower or the like is kept stably and constantly, and measured with a gas meter. Besides, calculate the flow rate of isokinetic sampling according to the formula (11) specified in 9.3.1 of the text of this Standard from the flow velocity of air measured with the Pitot tube simultaneously jointly set and the measured values of various conditions of air necessary for calculation of the flow rate of isokinetic sampling, and obtain the relative error of isokinetic sampling according to the formula (3) of 3.2.2 (2).
- (b) Balanced-type dynamic pressure-type sampling apparatus Suck air indoors for specific time, measure the suction amount of air and the suction time with a gas meter, and simultaneously measure the differential pressure of a Venturi-tube. Then, calculate the suction flow rate from the suction amount of air and the suction time, and obtain the flow velocity of suction air in nozzle according to the formula (1) of 3.2.2 (1). Besides, calculate the flow velocity of air by the specification of 7.3 of the text of this Standard by considering the same

dynamic pressure as the differential pressure of the Venturi tube is obtainable with a Pitot tube, and obtain the relative error of isokinetic sampling according to the formula (2) of 3.2.2 (1).

Remarks: In the tests of 3.2.2 and 3.2.3, obtain the density of air necessary for calculation of the flow velocity with a Pitot tube according to the formula (4).

$$\rho = \frac{1.293 \left(P_a - \frac{0.378 \varphi}{100} \cdot P_v \right)}{\left(1 + \frac{\theta_a}{273} \right) 101.3} \dots \dots \dots (4)$$

$$\left\{ \gamma = \frac{1.293 \left(P_a - \frac{0.378 \varphi}{100} \cdot P_v \right)}{\left(1 + \frac{\theta_a}{273} \right) 760} \right\}$$

where, ρ : density of air (kg/m³)
 $\{\gamma$: weight of air per unit volume (kgf/m³)
 P_a : atmospheric pressure (kPa){mmHg}
 φ : relative humidity (%)
 θ_a : temperature of atmosphere (°C)
 P_v : saturated water vapor pressure at θ_a °C
(kPa){mmHg}

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